



Negative Sampling in Recommendation: What makes a Good Negative?

MINDS Lab.

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2026-09-08

Contents

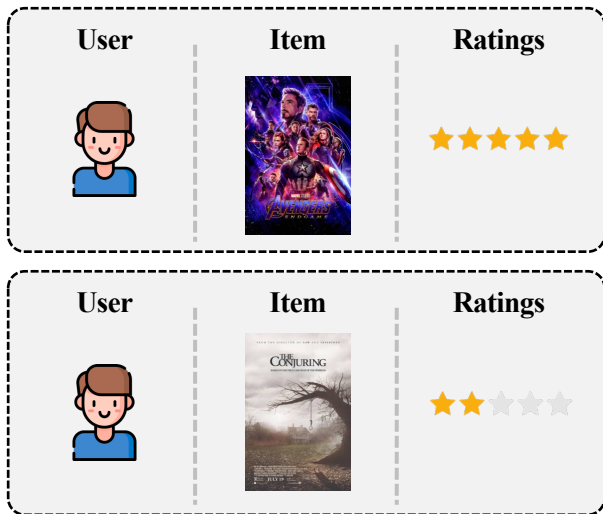
- ❖ Negative Sampling in Recommendation
- ❖ DNS: Mine Hard Negatives
- ❖ SRNS: Hard but Reliable Negatives
- ❖ DNS+: Why Does Hard Negative Sampling Work?

Recommendation Learns from User Feedback

Negative Sampling in Recommendation

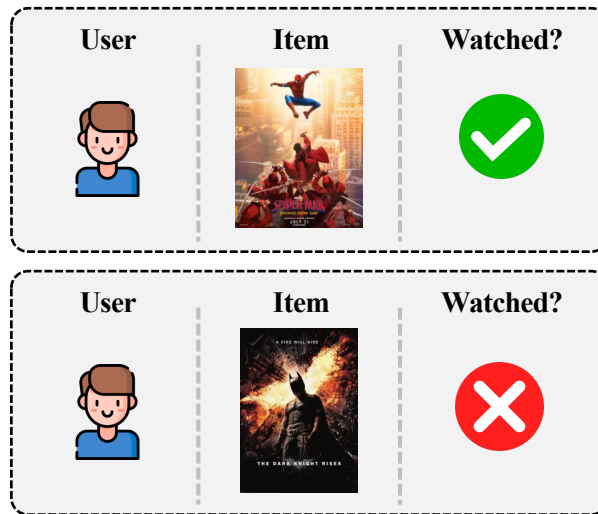
❖ Explicit Feedback

- ❑ Users directly express their preferences
 - Rating, Review, Like / Dislike, ...



❖ Implicit Feedback

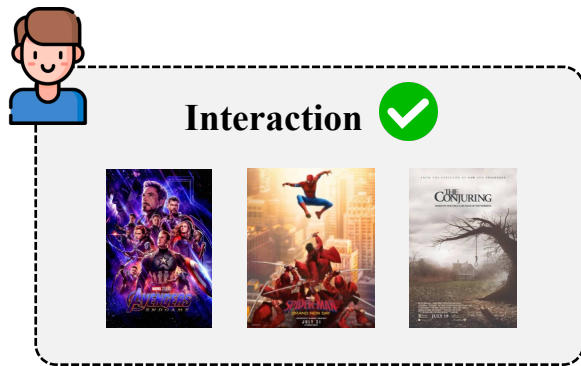
- ❑ Preferences are inferred from user behavior
 - Click, Purchase, Watch, Play, ...



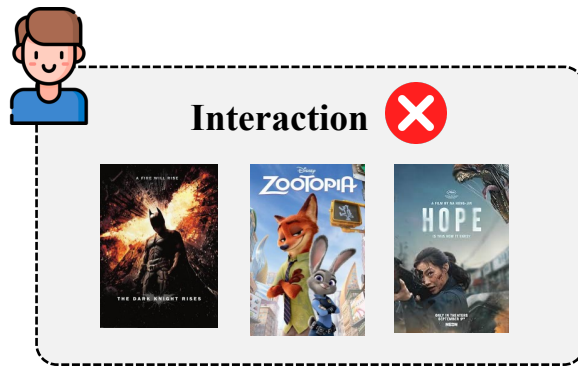
Negative Samples under Implicit Feedback

Negative Sampling in Recommendation

❖ Where do negative samples come from?



Observed → *Positive*



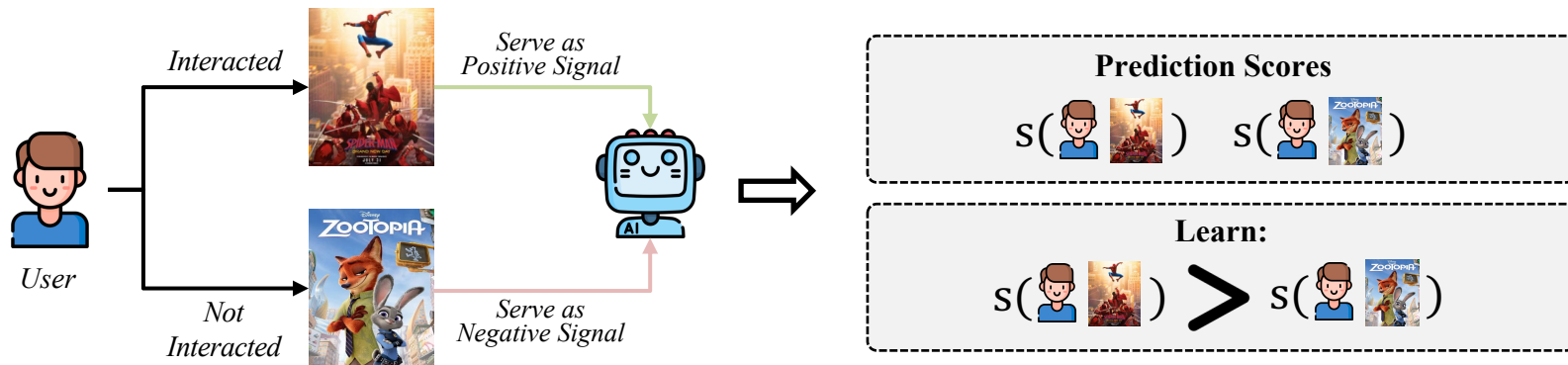
Unobserved →
Negative Candidate Pool

- ❑ Negative sampling selects a subset of training negatives from the massive unobserved item pool

How Are Positive and Negative Samples Used in Training?

Negative Sampling in Recommendation

- ❖ Learning relative preference from positive and negative samples

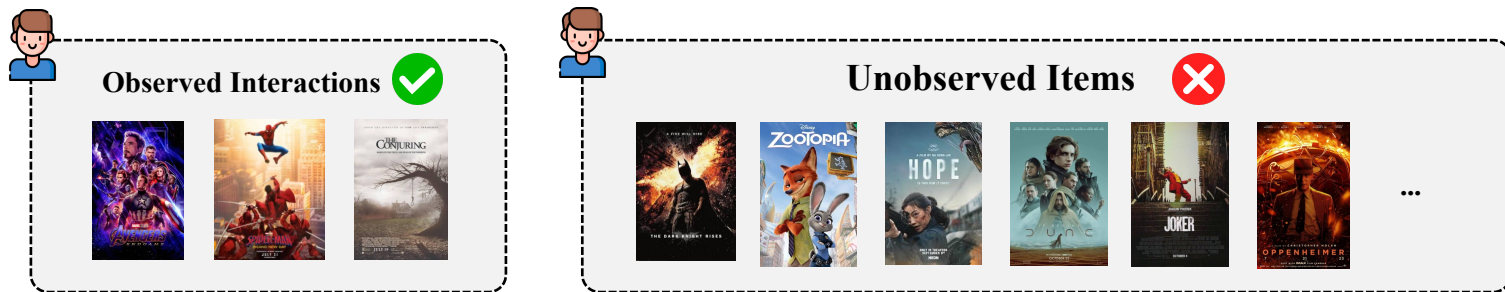


- ❑ Positive–Negative comparison
 - Preference scores for positive and sampled negative items
- ❑ Relative preference optimization
 - Higher score for positives than sampled negatives

Why Do We Need Sampling?

Negative Sampling in Recommendation

❖ Too many negative candidates for training



❑ Large-scale recommendation

- ▶ Millions of users & items
- ▶ Massive item corpus

❑ Sparse user interactions

- ▶ Few observed interactions per user
- ▶ Most user-item pairs remain unobserved

❑ Computational burden

- ▶ Full-corpus training is impractical
- ▶ High cost per training update

Uniform Negative Sampling

Negative Sampling in Recommendation

❖ The simplest sampling strategy



- ❑ Model-agnostic selection
 - No consideration of current model predictions
- ❑ Simple and efficient
 - No additional scoring or side information

From Hard Negatives to Better Negatives

Negative Sampling in Recommendation

❖ How the definition of a “good negative” evolves

- ❑ DNS — Find Hard Negatives
 - **Good Negative = Hardness**

- ❑ SRNS — Make Hard Negatives Reliable
 - **Good Negative = Hardness + Reliable**

- ❑ DNS+ — Control the Right Hardness
 - **Good Negative = Appropriately Hardness**



Optimizing Top-N Collaborative Filtering via Dynamic Negative Item Sampling

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Published in SIGIR'13

Do All Negatives Look the Same to the Current Model?

DNS: Mine Hard Negatives

❖ Same sampling rule, different model responses



❑ Different levels of separation

- Same sampling probability, but different score gaps from the positive

➔ Does this difference matter during training?

Why Does the Negative Score Matter?

DNS: Mine Hard Negatives

❖ Different score gaps, different optimization effects

$$\mathcal{L}_{BPR} = -\log \sigma(s(u, i^+) - s(u, j^-)), \quad s(u, i^+): 0.82$$

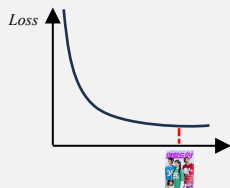


Well-separated Negative



$$s(u, j_1) = 0.05$$

$$\Delta_1 = 0.82 - 0.05 = 0.77$$



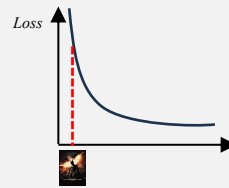
Smaller gradient magnitude

Poorly-separated Negative



$$s(u, j_2) = 0.78$$

$$\Delta_2 = 0.82 - 0.78 = 0.04$$



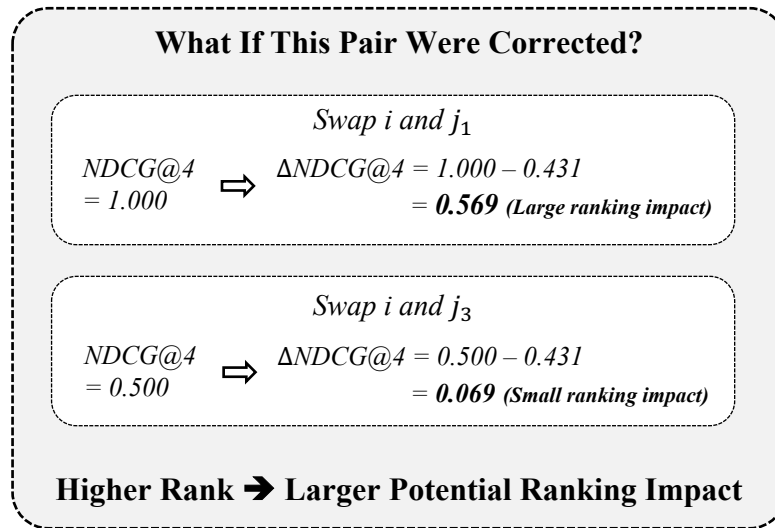
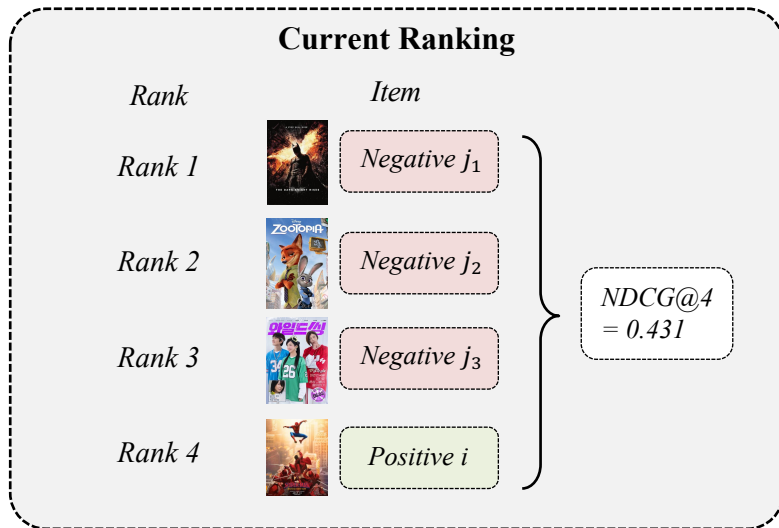
Larger gradient magnitude

- ❑ Already well-separated negatives provide less additional optimization pressure
- ❑ The usefulness of a negative depends on the current model state

Rank Matters

DNS: Mine Hard Negatives

- ❖ Top-N recommendation is evaluated by rank-biased metrics

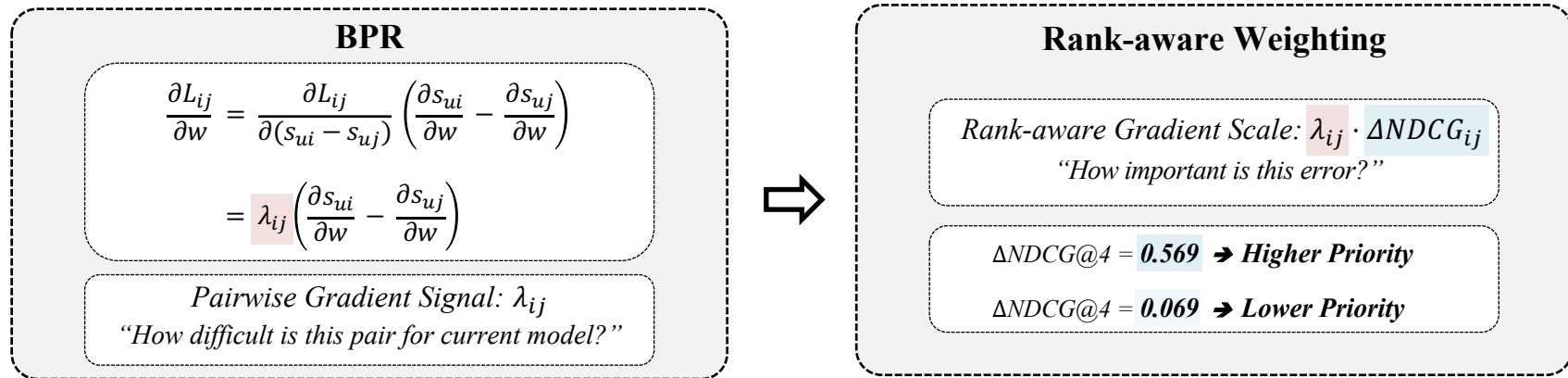


- ❑ Errors near the top matter more for rank-biased metrics

Weight What Matters

DNS: Mine Hard Negatives

❖ From ranking impact to learning priority



❑ But full ranking is expensive

- Computing $\Delta NDCG$ requires the current ranking over all unseen items

➔ Can we get the same effect through sampling instead of explicit weighting?

Sample What Matters

DNS: Mine Hard Negatives

❖ From rank-aware weighting to rank-aware sampling

Sampling Instead of Weighting

$$p_j \propto \frac{f(\lambda_{ij}, \zeta_u)}{\lambda_{ij}}, f(\lambda_{ij}, \zeta_u) = \lambda_{ij} \cdot \Delta NDCG_{ij} \\ \rightarrow p_j \propto \Delta NDCG_{ij}$$

Larger ranking impact $\rightarrow$ Higher sampling probability
But $\Delta NDCG$ itself still requires full ranking...



Rank-aware Reject Sampling (for one sample)

Step 1. Random Candidates



Step 2. Current Model Scores



Step 3. Pick a High-ranked Candidate

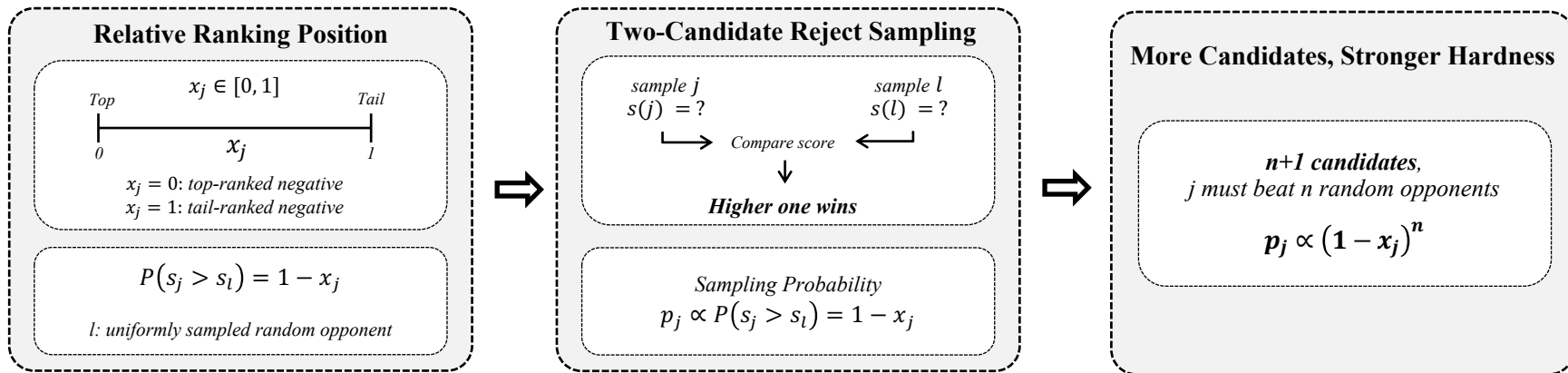


- Sample a few $\rightarrow$ Score them $\rightarrow$ Prefer higher-ranked negatives
 - Dynamic: the hard negative changes as the model changes

Sample What Matters (Cont.)

DNS: Mine Hard Negatives

❖ Why does reject sampling become rank-aware?



- ❑ Higher-ranked negatives win random comparisons more often → Higher Sampling Probability
- ❑ Random competition implicitly approximates a rank-biased sampling distribution

Does DNS Improve Learning?

DNS: Mine Hard Negatives

- ❖ DNS improves both ranking quality and training efficiency

Ranking Performance

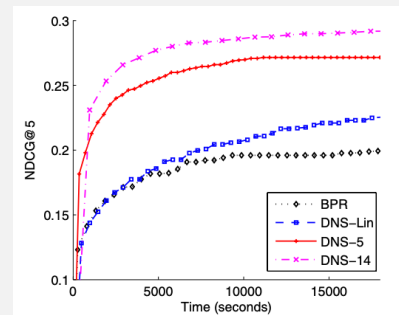
Netflix					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.3826	0.3272	0.2052	0.2017	0.1403
DNS	0.4708	0.4012	0.2906	0.2887	0.2036
Impv.	23.1%*	22.6%*	41.6%*	43.1%*	45.1%*

Yahoo! Music					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.1588	0.1359	0.1683	0.1481	0.0615
DNS	0.4243	0.3671	0.4458	0.3981	0.1644
Impv.	167.2%*	170.1%*	164.9%*	168.8%*	167.3%*

Last.fm					
	P@5	P@10	NDCG@5	NDCG@10	MAP
BPR	0.1231	0.1168	0.1270	0.1207	0.0221
DNS	0.1323	0.1202	0.1355	0.1250	0.0223
Impv.	7.5%*	2.9%	6.7%*	3.6%	0.9%

Harder negatives substantially improve rank-biased metrics

Training Efficiency



More informative updates require fewer training rounds

- ❑ Current model score provides a useful signal for negative selection
- ❑ Sampling informative negatives improves both ranking quality and convergence



Simplify and Robustify Negative Sampling for Implicit Collaborative Filtering

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Tsinghua University

Quanming Yao
4Paradigm Inc (Hong Kong)

Yong Li
Tsinghua University

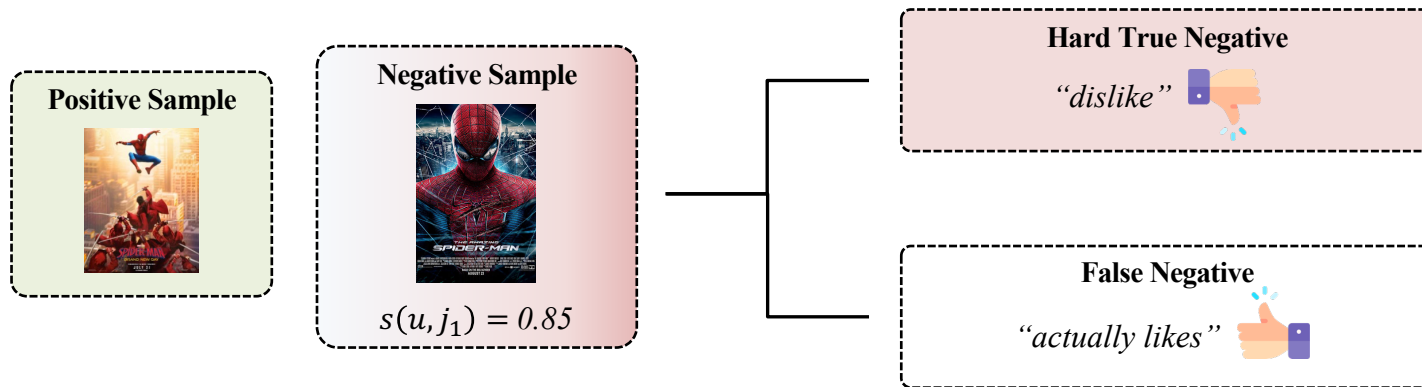
Depeng Jin
Tsinghua University

Published in NeurIPS'20

Is Hard Always Good?

SRNS: Hard but Reliable Negatives

❖ Hard negative sampling may amplify false negatives



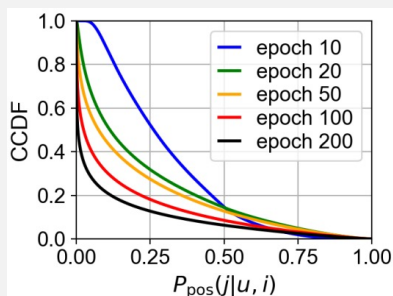
- ❑ An item may be unobserved because of exposure, position, or other factors — not necessarily dislike
 - High Score $\neq$ True Negative
- ❑ Can we distinguish hard true negatives from false negatives?

Understanding False Negatives

SRNS: Hard but Reliable Negatives

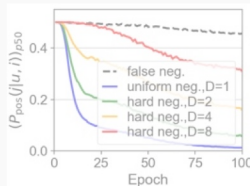
❖ Finding 1: Hard negatives are rare

Only a Few Negatives Are Hard



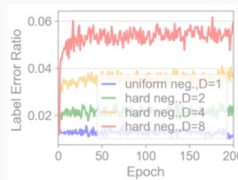
$$F(x) = P(P_{pos} \geq x)$$

False Negatives Also Look Hard



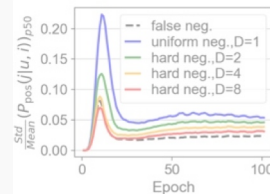
D : number of random candidates used to select one negative

Harder Sampling Selects More False Negatives



$$\text{Label Error Ratio} = \frac{\# \text{ selected false negatives}}{\# \text{ selected negatives}}$$

False Negatives Are More Stable



$$\text{Normalized Variance} = \frac{\text{Std}(P_{pos})}{\text{Mean}(P_{pos})}$$

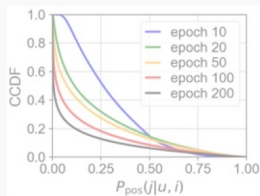
- ❑ Only a small fraction of unobserved items have high positive-label probability P_{pos}

Understanding False Negatives

SRNS: Hard but Reliable Negatives

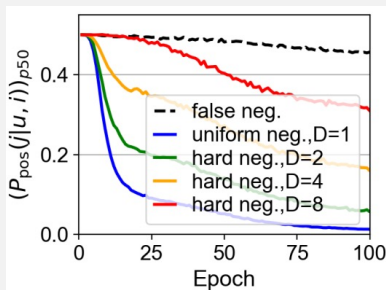
❖ Finding 2: Score captures hardness, but not reliability

Only a Few Negatives Are Hard



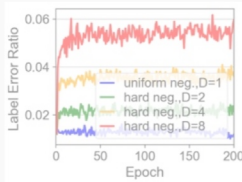
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False Negatives Also Look Hard



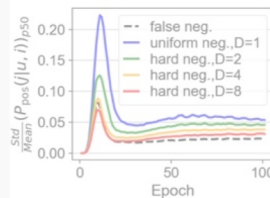
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Harder Sampling Selects More False Negatives



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False Negatives Are More Stable



$$\text{Normalized Variance} = \frac{\text{Std}(P_{pos})}{\text{Mean}(P_{pos})}$$

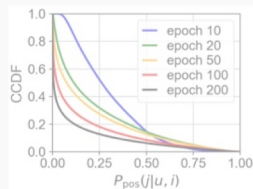
- ❑ Harder negatives have larger P_{pos} , but false negatives also maintain high scores
- ❑ Prediction score alone cannot distinguish hard true negatives from false negatives

Understanding False Negatives

SRNS: Hard but Reliable Negatives

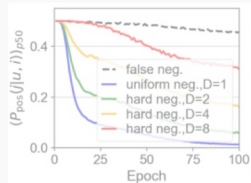
- ❖ Harder is more informative, but also noisier

Only a Few Negatives Are Hard



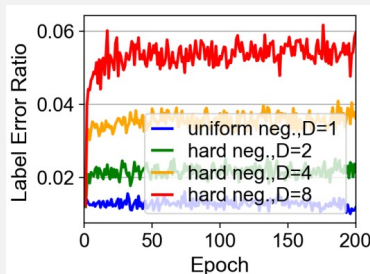
$$F(x) = P(P_{pos} \geq x)$$

False Negatives Also Look Hard



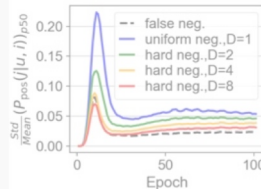
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Harder Sampling Selects More False Negatives



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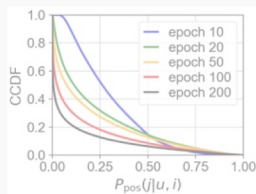
- ❑ As D increases, the Label Error Ratio consistently increases
 - ▶ Aggressive hard-negative mining increases false-negative contamination

Understanding False Negatives

SRNS: Hard but Reliable Negatives

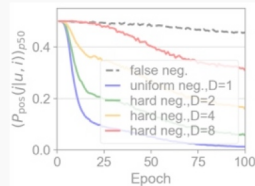
- ❖ Prediction variance provides an additional reliability signal

Only a Few Negatives Are Hard



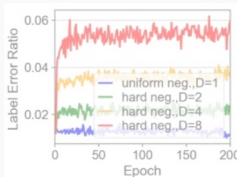
$$F(x) = P(P_{pos} \geq x)$$

False Negatives Also Look Hard



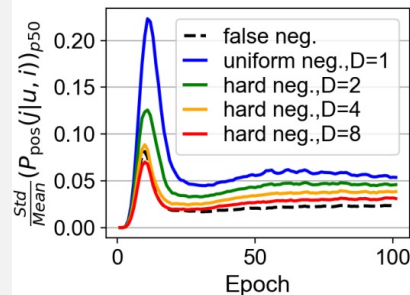
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Harder Sampling Selects More False Negatives



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False Negatives Are More Stable



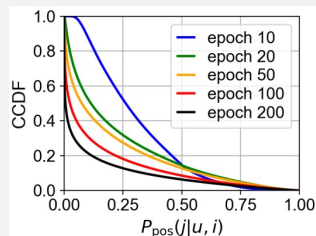
$$\text{Normalized Variance} = \frac{\text{Std}(P_{pos})}{\text{Mean}(P_{pos})}$$

- ❑ False negatives show lower prediction variance than true negative samples
 - Their predictions remain relatively stable throughout training

Understanding False Negatives (Cont.)

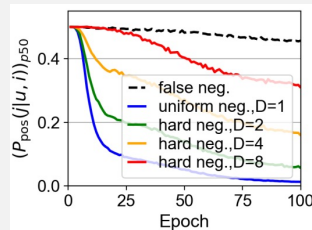
SRNS: Hard but Reliable Negatives

Only a Few Negatives Are Hard



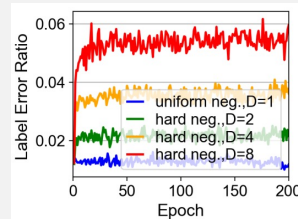
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False Negatives Also Look Hard



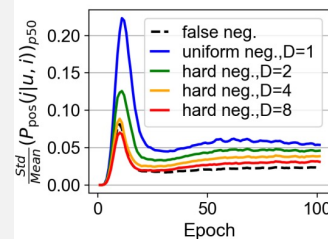
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False Negatives Are More Stable



$$\text{Normalized Variance} = \frac{\text{Std}(P_{pos})}{\text{Mean}(P_{pos})}$$

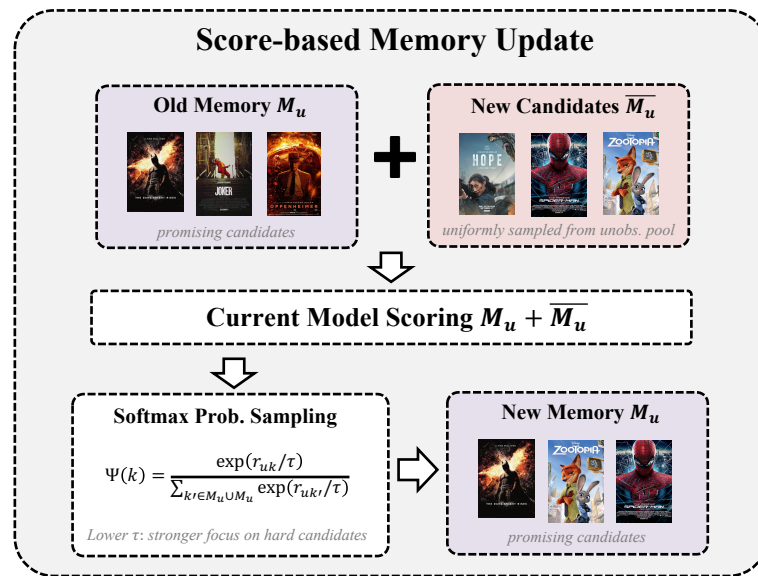
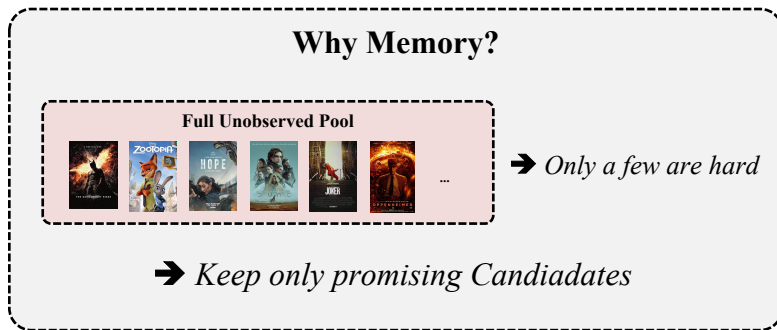
Current Score $\rightarrow$ Hardness

Prediction History $\rightarrow$ Reliability

Remember the Hard Ones

SRNS: Hard but Reliable Negatives

❖ Maintain a small memory of promising negatives

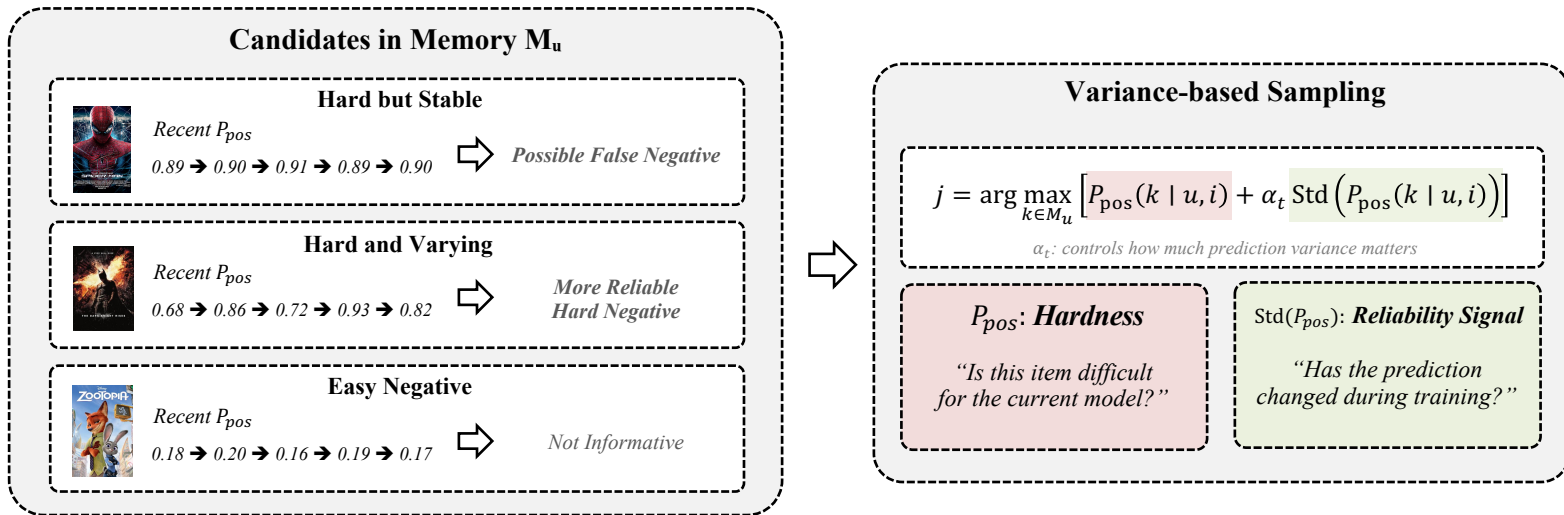


- ❑ Remember promising negatives instead of rediscovering them from the full pool
- ❑ But high-score memory can still contain false negatives. How should we choose one reliably?

Hard, But Reliable

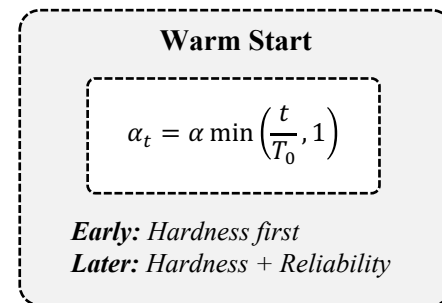
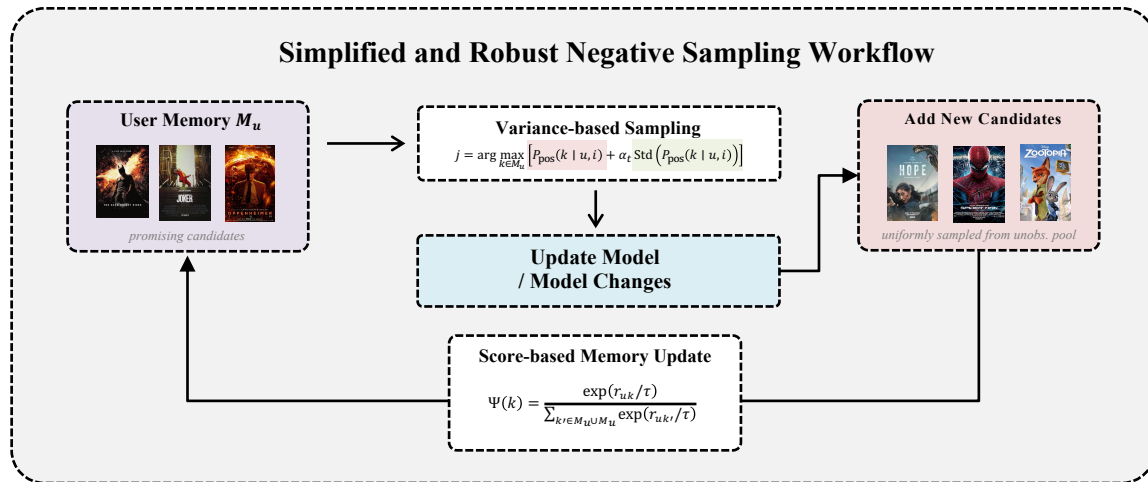
SRNS: Hard but Reliable Negatives

- ❖ Select negatives using both hardness and prediction stability



- ❑ Do not simply choose the hardest negative — prefer one that is hard and less likely to be false.

❖ Alternating between robust sampling and dynamic memory update

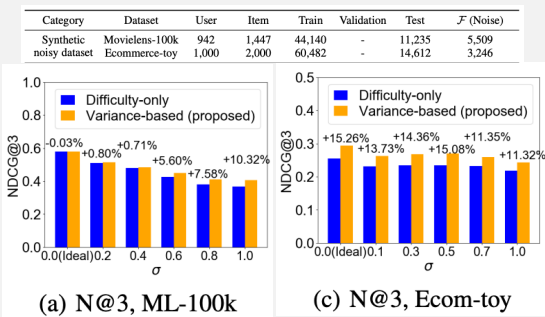


Does Reliability Help?

SRNS: Hard but Reliable Negatives

- ❖ Variance-aware sampling improves robustness and recommendation quality

Robustness to False Negatives



Variance-based sampling remains more robust as false-negative noise increases

Real-world Performance

Method	Movielens-1m			Pinterest			Ecommerce		
	N@1	N@3	R@3	N@1	N@3	R@3	N@1	N@3	R@3
ENMF	0.1846	0.3021	0.3882	0.2594	0.4144	0.5284	0.1317	0.2095	0.2670
Uniform	0.1744	0.2846	0.3663	0.2586	0.4136	0.5276	0.1265	0.2057	0.2640
NNCF	0.0829	0.1478	0.1971	0.2292	0.3699	0.4735	0.0833	0.1420	0.1855
AOBPR	0.1802	0.2905	0.3728	0.2596	0.4165	0.5319	0.1293	0.2108	0.2710
IRGAN	0.1755	0.2877	0.3708	0.2587	0.4143	0.5282	0.1275	0.2065	0.2648
RNS-AS	0.1823	0.2932	0.3754	0.2690	0.4233	0.5359	0.1335	0.2131	0.2714
AdvIR	0.1790	0.2941	0.3792	0.2689	0.4235	0.5363	0.1357	0.2141	0.2719
SRNS	0.1933	0.3070	0.3912	0.2891	0.4391	0.5486	0.1471	0.2256	0.2833
	4.71%	1.62%	0.77%	7.47%	3.68%	2.29%	8.40%	5.37%	4.19%

- ❑ Current score finds informative negatives, Prediction history reduces the risk of false negatives



On the Theories Behind Hard Negative Sampling for Recommendation

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University of Science and
Technology of China

Jiawei Chen

Zhejiang University

Fuli Feng

University of Science and
Technology of China

Jizhi Zhang

University of Science and
Technology of China

Junkang Wu

University of Science and
Technology of China

Chongming Gao

University of Science and
Technology of China

Xiangnan He

University of Science and
Technology of China

Published in WWW'23

Beyond Faster Convergence

DNS+: Why Does Hard Negative Sampling Work?

❖ Is faster convergence enough to explain HNS?

Conventional view

*Hard Negative → Larger Gradient
→ Faster Convergence*

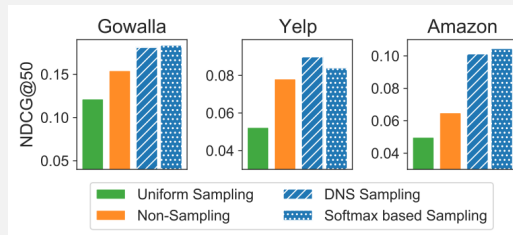
*This explains training efficiency —
but does it explain better final accuracy?*

BPR Loss with Sampling Distribution

$$\min_{\theta} \sum_{c \in \mathcal{C}} \sum_{i \in \mathcal{I}_c^+} E_{j \sim P_{ns}(j|c)} [\ell(r(c, i|\theta) - r(c, j|\theta))]$$

*Negative Sampling Distribution changes
the effective training objective*

An Unexpected Observation



HNS > Non-Sampling > Uniform

But Non-sampling uses ALL negatives

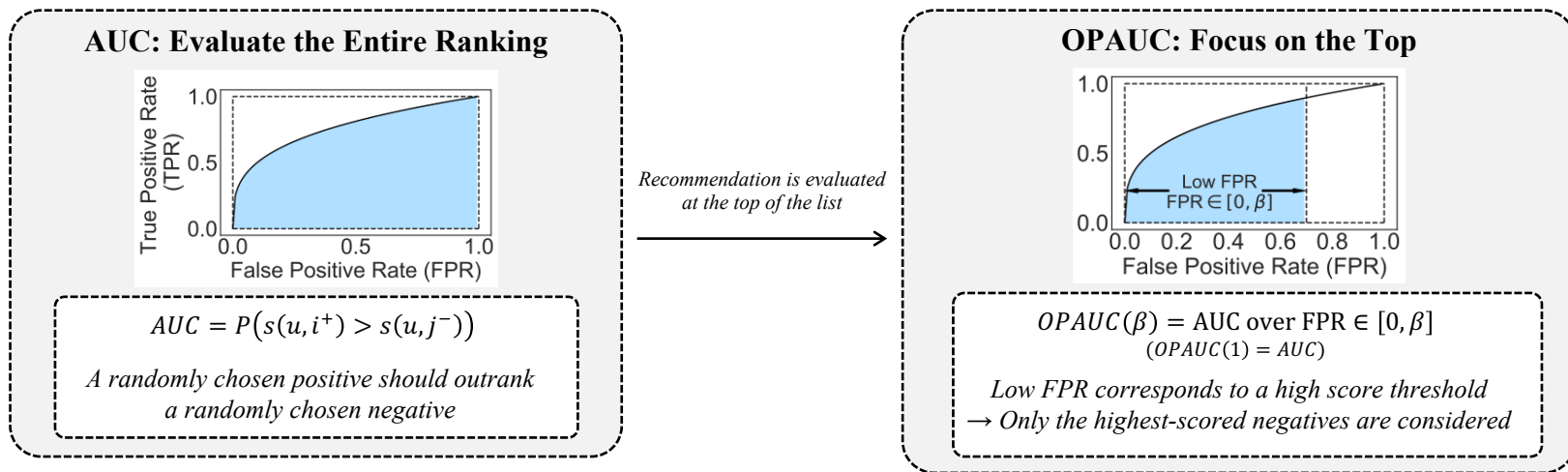
□ HNS may change not only how fast we optimize, but what we optimize

➔ What does Hard Negative Sampling actually optimize?

Focus on the Top

DNS+: Why Does Hard Negative Sampling Work?

- ❖ Top-K recommendation does not care about every ranking error equally

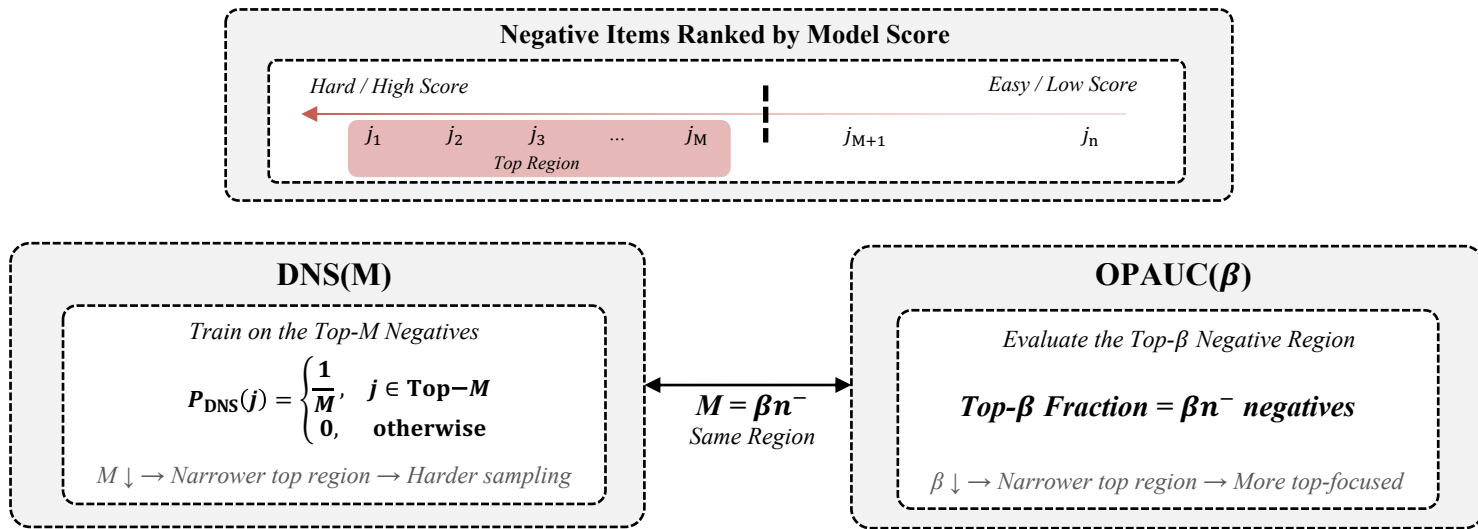


- AUC considers the whole negative population, while OPAUC focuses on top-ranked high-scored negatives

Why Does HNS Optimize OPAUC?

DNS+: Why Does Hard Negative Sampling Work?

- ❖ Both DNS and OPAUC focus on the same top-ranked negative region



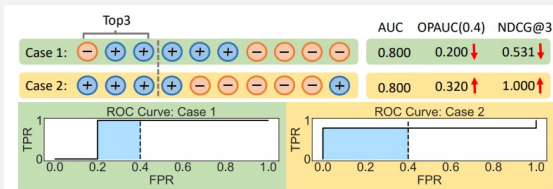
- ❑ BPR + DNS $\equiv$ OPAUC(β) surrogate (Formal proof via CVaR-DRO)
- ❑ DNS shifts learning toward the ranking region that matters most at the top

Why OPAUC Matches Top-K

DNS+: Why Does Hard Negative Sampling Work?

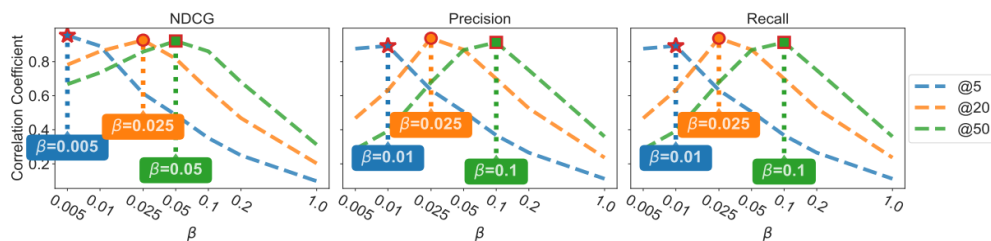
- ❖ The same overall ranking quality can produce very different Top-K results

Same AUC, Different Top-K



AUC may miss where ranking errors occur, while OPAUC is sensitive to errors near the top

Correlation with Top-K Metrics



OPAUC achieves stronger correlation with NDCG, Precision, and Recall than AUC

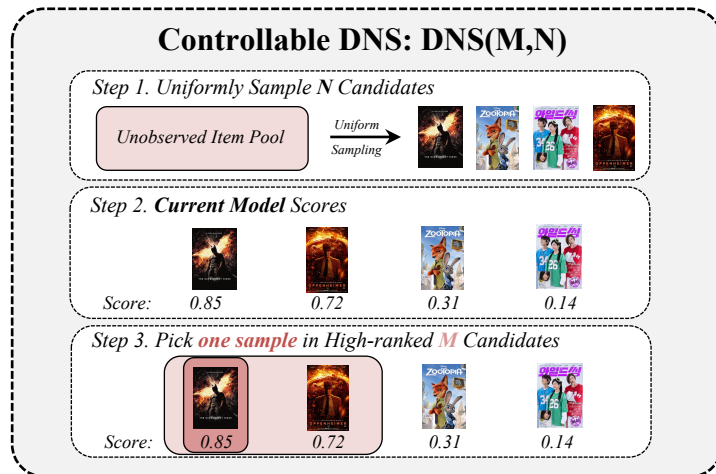
$K \downarrow \rightarrow \beta \downarrow$
Smaller K favors more top-focused objective

- ❑ AUC measures global pairwise ranking quality, regardless of where the errors occur
- ❑ OPAUC emphasizes errors among high-ranked negatives
 - which are exactly the errors that affect Top-K recommendation
- ❑ As the target K becomes smaller, the optimal OPAUC region also becomes narrower

How Hard Should It Be?

DNS+: Why Does Hard Negative Sampling Work?

- ❖ Optimal hardness depends on the target Top-K metric



Top-K Determines the Right Hardness

$K \downarrow \rightarrow \beta \downarrow$
 $\rightarrow$ Harder Negatives

- ❑ Hardness should be tuned to the ranking region emphasized by the target metric
- ❑ Smaller K focuses evaluation closer to the top, requiring harder negative samples

Does Controlled Hardness Help?

DNS+: Why Does Hard Negative Sampling Work?

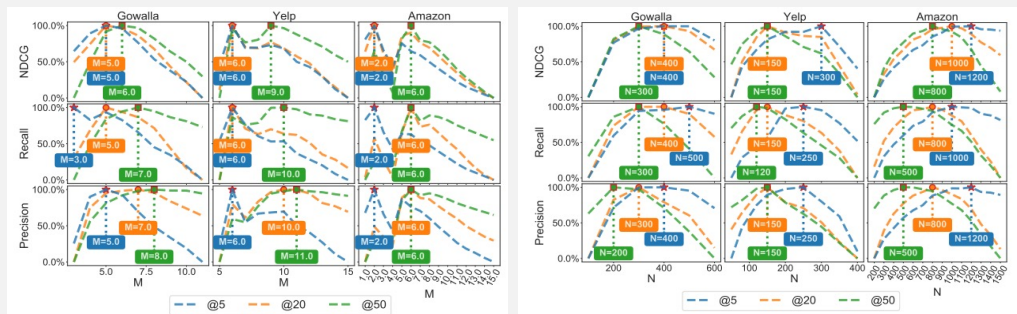
- ❖ Controllable hardness improves performance and follows the Top-K guideline

Controllable HNS Improves Performance

Method	Gowalla		Yelp		Amazon	
	NDCG@50	Recall@50	NDCG@50	Recall@50	NDCG@50	Recall@50
BPR	0.1216	0.2048	0.0524	0.1083	0.0499	0.1171
AOBPR	0.1385	0.2417	0.0677	0.1346	0.0563	0.1303
WARP	0.1248	0.2240	0.0636	0.1332	0.0542	0.1267
IRGAN	0.1443	0.2242	0.0695	0.1367	0.0627	0.1395
Kernel	0.1399	0.2264	0.0658	0.1315	0.0700	0.1495
DNS	0.1412	0.1839	0.0693	0.1425	0.0615	0.1378
PRIS(U)	0.1334	0.2217	0.0639	0.1273	0.0607	0.1377
PRIS(F)	0.1385	0.2282	0.0673	0.1342	0.0697	0.1463
AdaSIR(U)	0.1489	0.2500	0.0732	0.1523	0.0731	0.1505
AdaSIR(P)	0.1519	0.2516	0.0731	0.1525	0.0740	0.1534
DNS(M, N)	0.1811**	0.2989**	0.0899**	0.1774**	0.1014**	0.1833**
Softmax-v(ρ , N)	0.1837**	0.2993**	0.0840**	0.1690**	0.1046**	0.1937**

Controlling hardness consistently outperforms fixed hard-negative sampling

Smaller K Prefers Harder Negatives



$K \downarrow \Rightarrow M^* \downarrow, N^* \uparrow$
 $\Rightarrow$ Harder Sampling

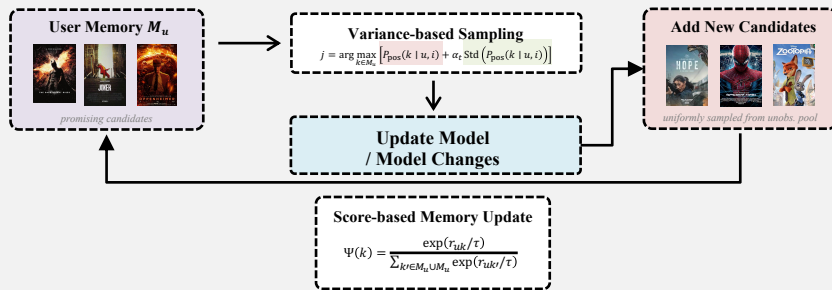
- ❑ Hardness should be controlled according to the ranking region emphasized by the target metric

Conclusion

Uniform Negative Sampling



Simplified and Robust Negative Sampling (SRNS)



Rank-aware Reject Sampling (DNS)

Step 1. Random Candidates



Step 2. Current Model Scores

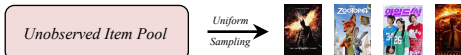


Step 3. Pick a High-ranked Candidate



Controllable DNS: DNS(M,N)

Step 1. Uniformly Sample N Candidates



Step 2. Current Model Scores



Step 3. Pick **one** sample in High-ranked **M** Candidates





Appendix (SRNS Time Complexity Table)

rank sorted by score, pop_j is the j 's item popularity, B is the mini-batch size, T is the time complexity of computing an instance score, E is the epoch of lazy-update, and $\mathcal{F}$ denotes false negative.

	$p_{\text{ns}}(j u)$	Optimization	Time Complexity	Robustness
Uniform [33]	Uniform($\{j \notin \mathcal{R}_u\}$)	SGD (from scratch)	$O(BT)$	×
NNCF [10]	$\propto (\text{pop}_j)^{0.75}$	SGD (from scratch)	$O(B^2T)$	×
AOBPR [32]	$\propto \exp(-\text{rk}(j u)/\lambda)$	SGD (from scratch)	$O(BT)$	×
IRGAN [37]	learned $\bar{p}_{\text{ns}}(j u)$ (GAN)	REINFORCE (pretrain)	$O(B \mathcal{I} T)$	×
AdvIR [28]	learned $\bar{p}_{\text{ns}}(j u)$ (GAN)	REINFORCE (pretrain)	$O(BS_1T)$	×
SRNS (proposed)	variance-based (see [4])	SGD (from scratch)	$O(\frac{B}{E}(S_1 + S_2)T)$	✓

DRO as a Bridge to Hard Negative Sampling

❖ DRO asks for the worst-case sampling distribution

□ Worst-case Reweighting

- $\max_Q \mathbb{E}_{j \sim Q} [L(c, i, j)] \quad \text{s. t.} \quad D_{\text{CVaR}}(Q \| P_0) \leq \rho$
 - Choose a negative-sampling distribution Q that maximizes the current loss, but cannot move too far from P_0 (Uniform Distribution)

□ CVaR constraint limits each probability

- $D_{\text{CVaR}}(Q \| P_0) = \sup_j \log \frac{Q(j)}{P_0(j)} \leq \rho, \quad Q(j) \leq e^\rho P_0(j) = \frac{e^\rho}{n_-}$

□ Then what is the worst-case Q^* ?

- $Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(L) \\ 0, & \text{otherwise} \end{cases}, \quad M = \frac{1}{e^\rho/n_-} = n_- e^{-\rho}$

DRO as a Bridge to Hard Negative Sampling (Cont.)

❖ Why is this DNS?

- BPR Loss: $L(c, i, j) = \log(1 + \exp(r_{cj} - r_{ci}))$, $r_{cj} \uparrow \Leftrightarrow L(c, i, j) \uparrow$
- Negative score ranking = BPR loss ranking $\rightarrow \text{TopM}(\text{ScoreRank}) = \text{TopM}(\text{Loss})$

❖ DRO $\leftrightarrow$ HNS (DNS)

- $Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(L) \\ 0, & \text{otherwise} \end{cases}$, $\text{TopM}(L) = \text{TopM}(r)$

- $\rightarrow Q^*(j) = \begin{cases} \frac{1}{M}, & j \in \text{TopM}(r) \\ 0, & \text{otherwise} \end{cases} = P_{\text{DNS}}(j)$

From OPAUC Metric to Trainable Objective

❖ How a partial ranking metric becomes a BPR-style loss

□
$$OPAUC(\beta) = \frac{1}{|C|} \sum_{c \in C} \int_0^\beta T PR_{c,\theta} [FPR_{c,\theta}^{-1}(s)] ds$$

- OPAUC only evaluates the low-FPR region

□ Convert it to a finite-data pairwise estimator

- $$\widehat{OPAUC}(\beta) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{n_+ n_-} \sum_{i \in \mathcal{I}_c^+} \sum_{j \in \mathcal{S}_{j_c^-}^{\downarrow} [1, n_- \beta]} \mathbb{I}(r_{ci} > r_{cj})$$
- negative items sorted by score, taking only the top $n_- \beta$

□ Replace the indicator with a surrogate loss

- $$L(c, i, j) = \ell(r_{ci} - r_{cj})$$
- $$\min_{\theta} \frac{1}{|C|} \sum_{c \in C} \frac{1}{n_+} \sum_{i \in \mathcal{I}_c^+} \frac{1}{n_- \beta} \sum_{j \in \mathcal{S}_{j_c^-}^{\downarrow} [1, n_- \beta]} L(c, i, j)$$

Why Does DRO Optimize OPAUC?

❖ OPAUC and CVaR-DRO optimize the same top-ranked negative region

□ OPAUC as a trainable objective

- $\mathcal{L}_{\text{OPAUC}}(\beta) = \frac{1}{n_- \beta} \sum_{j \in \text{Top}_{n_- \beta}} L(c, i, j)$
- Average BPR-style loss over only the Top- β fraction of negatives

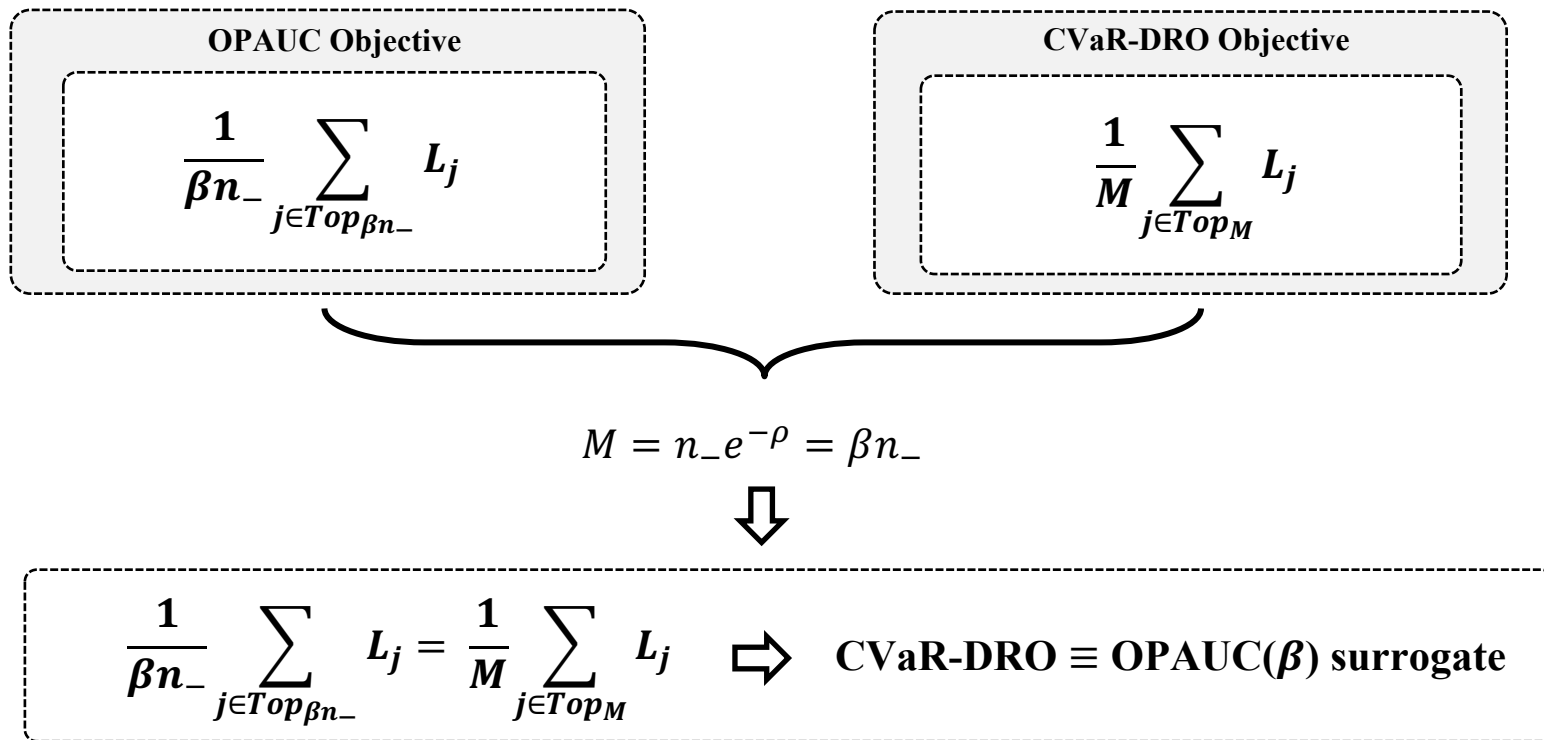
□ Recall the CVaR-DRO

- $\mathcal{L}_{\text{DRO}} = \frac{1}{M} \sum_{j \in \text{Top}_M} L(c, i, j)$

□ LEMMA: $\beta = e^{-\rho}$

- $M = n_- e^{-\rho} = n_- \beta$

Why Does DRO Optimize OPAUC? (Cont.)



Why Does OPAUC Connect to Top-K?

Top-K Quality

$$\text{Recall@K} = \frac{m}{N_+}, \text{ Precision@K} = \frac{m}{K}$$

$m = \# \text{positives in Top-K}$

Fix m, Find Extremes

Best arrangement

$$\underbrace{+\cdots+}_m \underbrace{-\cdots-}_{K-m} \mid \underbrace{+\cdots+}_{N_+=m} \underbrace{-\cdots-}_{N_-=K+m}$$

$\text{OPAUC}_{\max}(m)$

Worst arrangement

$$\underbrace{-\cdots-}_{K-m} \underbrace{+\cdots+}_m \mid \underbrace{-\cdots-}_m$$

$\text{OPAUC}_{\min}(m)$



$$\text{OPAUC}_{\min}(m) \leq A \leq \text{OPAUC}_{\max}(m)$$

A : Actual ranking's OPAUC

Solve for m
under fixed A

$$m_{\min}(A) \leq m \leq m_{\max}(A)$$



$$\frac{m_{\min}}{N_+} \leq \text{Recall@K} \leq \frac{m_{\max}}{N_+}$$

$$\frac{m_{\min}}{K} \leq \text{Precision@K} \leq \frac{m_{\max}}{K}$$